

COURSE : STIQK2113

DISCRETE STRUCTURE

4.3 EQUIVALENCE RELATIONS
4.4 OPERATIONS ON RELATIONS
4.5 TRANSITIVE CLOSURE

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4.3 EQUIVALENCE RELATIONS

Equivalence Relations

- A relation R on a set A is called equivalence relation if it is:
 - reflexive,
 - symmetric and
 - transitive

Example 1

Let $A = \{a, b, c\}$. Determine whether the relation R whose matrix M_R is given is an equivalence relation. Justify your answer.

Solution:

check **a) Reflexive** : have all 1's in its main diagonal

b) Symmetric : $M_{12}=0$ then $M_{21}=0$ or
 $M_{13}=0$ then $M_{31}=0$ or
 $M_{23}=1$ then $M_{32}=1$ or

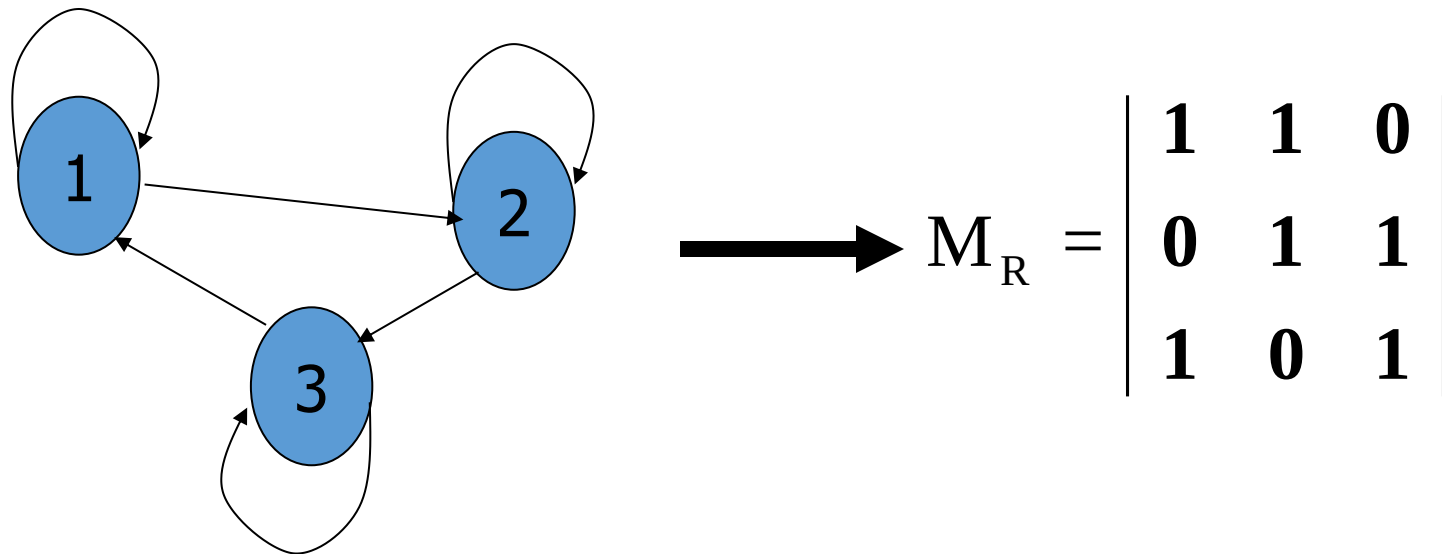
c) Transitive : $(M_R)^2_{\odot} = M_R$

So, R is an equivalence relation.

$$M_R = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix}$$

Example 2

Let $A = \{1, 2, 3\}$. Determine whether the relation R whose digraph is given is an equivalence relation. Justify your answer.



Example 2 (continued)

Solution:

check a) **Reflexive** :have all 1's in its main diagonal

b) **Not symmetric** : $M_{12}=1$ then $M_{21}=0$

$M_{13}=0$ then $M_{31}=1$

$M_{23}=1$ then $M_{32}=0$

c) **Not transitive** :

$$M_R = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix}$$

} violates

$$(M_R)^2_{\theta} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} \neq M_R$$

So, R is NOT an equivalence relation.

4.4 OPERATIONS ON RELATIONS

Complementary and Inverse

Let R and S be relations from a set A to a set B . It means R and S are subsets of $A \times B$ and we can use set operations on R and S , such as:

1. Complementary of $\bar{R}$

- **Definition:** $a \bar{R} b$ if and only if $a \not R b$
- **Explanation:** The complement of a relation R contains all the pairs that are **not in** R .
Formally, if $R \subseteq A \times A$, then: $\bar{R} = (A \times A) \setminus R$ (“ $\setminus$ ” set minus)

2. Inverse of R^{-1}

- **Definition:** $b R^{-1} a$ if and only if $a R b$
- **Explanation:** The inverse of a relation R consists of all pairs with the elements flipped.
- If $(a, b) \in R$, then $(b, a) \in R^{-1}$

Union and Intersection

3. Union of R and S : $R \cup S$

Definition: The union of relations R and S includes all ordered pairs that are in either R or S or both.

4. Intersection of R and S : $R \cap S$

Definition: The intersection of relations R and S includes all ordered pairs that are in both R and S .

Matrix Representation

1. Complement of Relation R

$$M_{\overline{R}} = \overline{M_R}$$

- This means the complement of the relation R is obtained by taking the **Boolean negation** of its matrix M_R .
- If $M_R[i][j] = 1$, then $\overline{M_R}[i][j] = 0$, and vice versa.

2. Inverse of Relation R

$$M_{R^{-1}} = [M_R]^T$$

- The inverse of the relation corresponds to the **transpose** of the matrix.
- So if R contains (a, b) , then R^{-1} contains (b, a) , which is just flipping the rows and columns.

Matrix Representation

3. Intersection of Relations $R \cap S$

$$M_{R \cap S} = M_R \wedge M_S$$

- The matrix of the intersection is the **element-wise Boolean AND** of the matrices M_R and M_S .
 - Only entries where both M_R and M_S are 1 will remain 1.
-

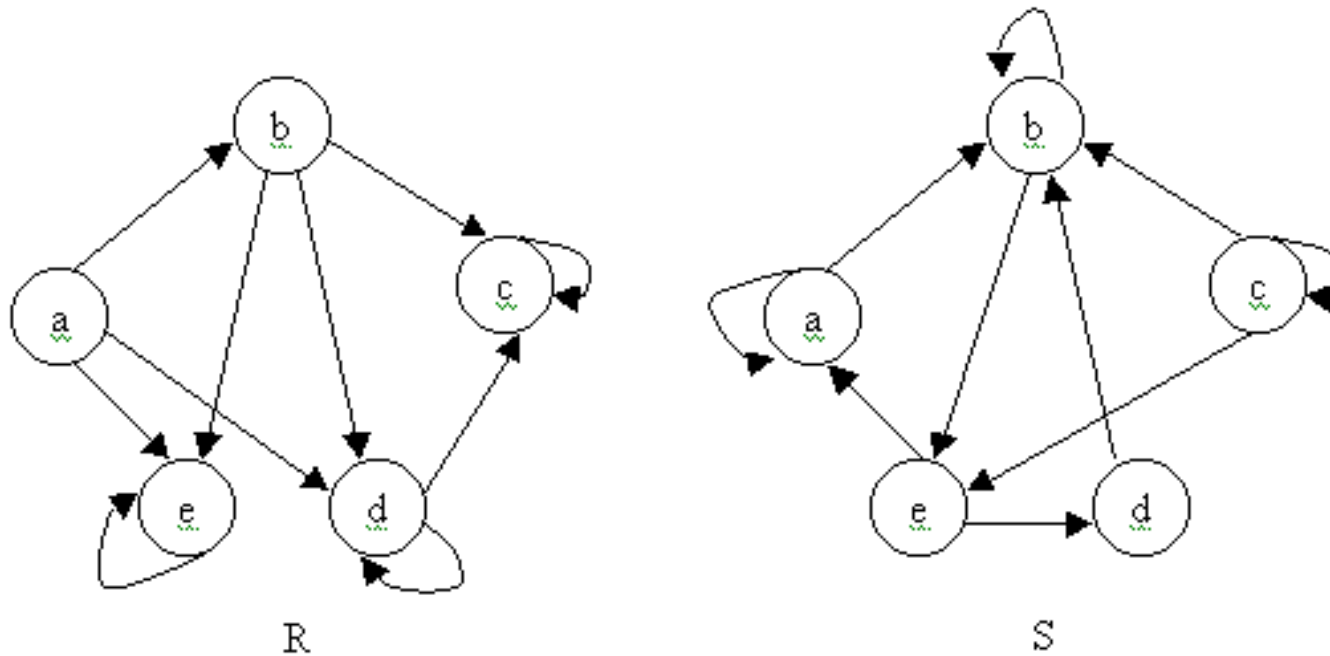
4. Union of Relations $R \cup S$

$$M_{R \cup S} = M_R \vee M_S$$

- The matrix of the union is the **element-wise Boolean OR** of M_R and M_S .
- If either matrix has a 1 at a position, the result will have a 1.

Example 1

Let R and S be two relations whose corresponding digraphs are shown below.



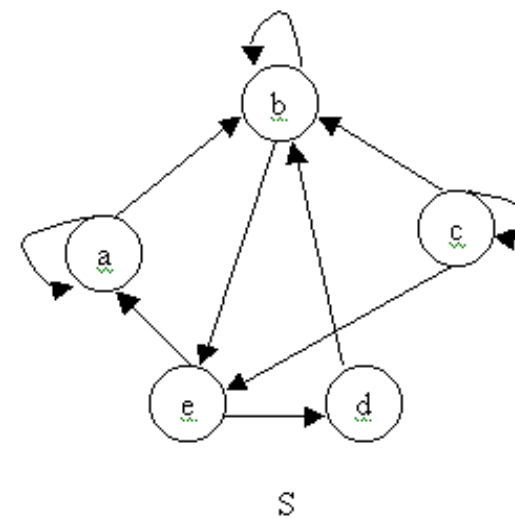
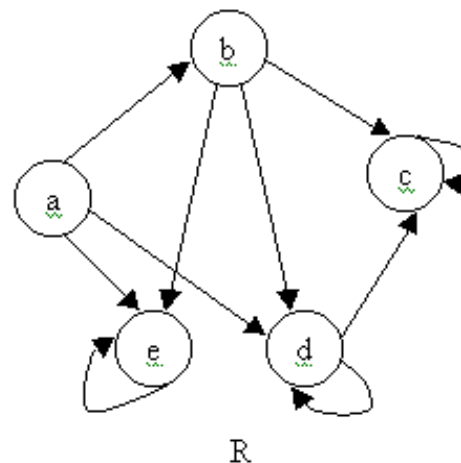
Find $a) \overline{R}$ $b) R \cap S$ $c) R \cup S$ $d) S^{-1}$

Example 1

Solution:

$R = \{(a,b), (a,d), (a,e), (b,c), (b,d), (b,e), (c,c), (d,c), (d,d), (e,e)\}$

$S = \{(a,a), (a,b), (b,b), (b,e), (c,b), (c,c), (c,e), (d,b), (e,a), (e,d)\}$



$$a) \bar{R} = \{(a,a), (a,c), (b,a), (b,b), (c,a), (c,b), (c,d), (c,e), (d,a), (d,b), (d,e), (e,a), (e,b), (e,c), (e,d)\}$$

$$b) R \cap S = \{(a,b), (b,e), (c,c)\}$$

$$c) R \cup S = \{(a,a), (a,b), (a,d), (a,e), (b,b), (b,c), (b,d), (b,e), (c,b), (c,c), (c,e), (d,b), (d,c), (d,d), (e,a), (e,d), (e,e)\}$$

$$d) S^{-1} = \{(a,a), (b,a), (b,b), (e,b), (b,c), (c,c), (e,c), (b,d), (a,e), (d,e)\}$$

Example 2

Let R and S be two relations whose corresponding matrices are shown below.

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Find a) $M_{\bar{R}}$ b) $M_{R \cap S}$ c) $M_{R \cup S}$ d) $M_{S^{-1}}$

Example 2

Solution:

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$a) M_{\bar{R}} = \overline{M_R} = \begin{bmatrix} \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} \end{bmatrix}$$

$$b) M_{R \cap S} = M_R \wedge M_S = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

$$c) M_{R \cup S} = M_R \vee M_S = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \end{bmatrix}$$

$$d) M_{S^{-1}} = (M_S)^T = \begin{bmatrix} \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

4.5 TRANSITIVE CLOSURE

Transitive Closure

- The transitive closure R^∞ of a relation R is the smallest transitive relation containing R.
- In simpler terms, it adds the minimum number of ordered pairs to R such that if aRb and bRc , then aRc is also included.

Three methods of finding R^∞ :

a) Digraph

b) Matrix

c) Warshall's Algorithm

Example

Let $A = \{1,2,3,4\}$, and

let $R = \{(1,2),(2,3),(3,4),(2,1)\}$.

Find the transitive closure of R .

Solution

Method 1: Digraph

- computing all path

Vertex 1: have paths to vertices 2,3,4 and 1

(1,2),(1,3),(1,4) and (1,1)

Vertex 2: have paths to vertices 1,3,4 and 2

(2,1),(2,3),(2,4) and (2,2)

Vertex 3: have paths to vertices 4

(3,4)

Vertex 4: no path

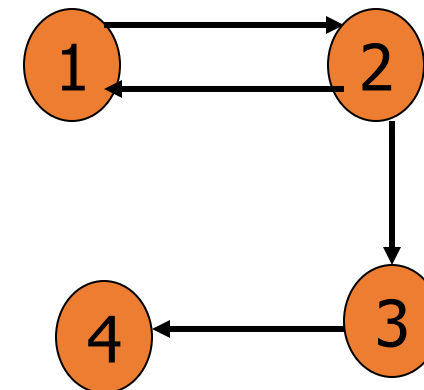
$$\therefore R^{\infty} = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,4)\}$$

Question

Let $A = \{1,2,3,4\}$, and

let $R = \{(1,2), (2,3), (3,4), (2,1)\}$.

Find the transitive closure of R .



ANSWER

Solution

Method 2: Matrix

Using formula:

$$M_{R^\infty} = M_R \vee (M_R)^2_\theta \vee (M_R)^3_\theta$$

convert R to M_R

$$R = \{(1,2),(2,3),(3,4),(2,1)\}$$



$$M_R = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

Find $(M_R)^2_\theta$ and $(M_R)^3_\theta$

Question

Let $A = \{1,2,3,4\}$, and

let $R = \{(1,2),(2,3),(3,4),(2,1)\}$.

Find the transitive closure of R.

Solution

$$(M_R)^2_{\theta} = M_R \Theta M_R$$

$$= \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Theta \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$(M_R)^3_{\theta} = (M_R)^2_{\theta} \Theta M_R$$

$$= \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \Theta \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

Replace $M_R, (M_R)^2_{\theta}$ and $(M_R)^3_{\theta}$ into the formula.

Solution

$$M_{R^\infty} = M_R \vee (M_R)^2_\theta \vee (M_R)^3_\theta$$

$$= \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix} \vee \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix} \vee \begin{vmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

$$\therefore R^\infty = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,4)\}$$

ANSWER

Solution

Question

Let $A = \{1,2,3,4\}$, and
let $R = \{(1,2),(2,3),(3,4),(2,1)\}$.
Find the transitive closure of R .

Method 3: Marshall's Algorithm

convert R to M_R

$$R = \{(1,2),(2,3),(3,4),(2,1)\}$$



$$M_R = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

Solution

Step 1:

$$W_0 = M_R = \begin{array}{c} \downarrow \\ \left| \begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right| \leftarrow \end{array}$$

Observe value 1's in column 1 : 2

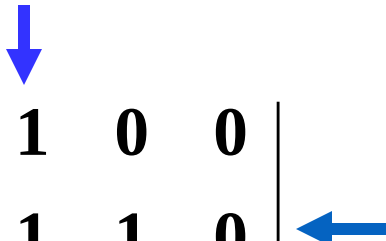
Observe value 1's in row 1 : 2

So we have new 1 in position (2,2).

$$W_1 = \begin{array}{c} \left| \begin{array}{cccc} 0 & 1 & 0 & 0 \\ 1 & \textcircled{1} & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right| \end{array}$$

Solution

Step 2:

$$W_1 = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$


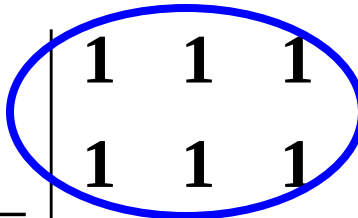
Observe value 1's in column 2 : 1,2

Observe value 1's in row 2 : 1,2,3

So we have new 1 in position:

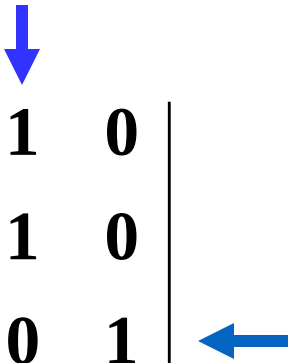
(1,1),(1,2),(1,3)(2,1),(2,2),(2,3)

(if 1's are not already there)

$$W_2 = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$


Solution

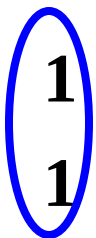
Step 3:

$$W_2 = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$


Observe value 1's in column 3 : 1,2

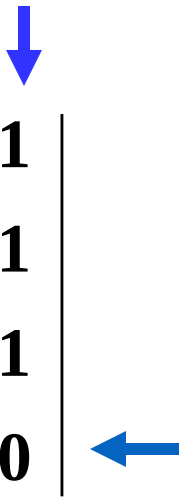
Observe value 1's in row 3 : 4

So we have new 1 in position: (1,4),(2,4)
(if 1's are not already there)

$$W_3 = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$


Solution

Step 4:

$$W_3 = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$


Observe value 1's in column 4 : **1,2,3**

Observe value 1's in row 4 : **N/A**

$$W_4 = W_3 = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

So we have **no new 1** are added.

$$\therefore R^\infty = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,4)\}$$



ANSWER

4.6 PARTIALLY ORDERED SET (POSET)

Partially Ordered Sets

A relation R on a set A is called partial order if R is:

- Reflexive,
- Antisymmetric, and
- Transitive

Example 1

Let $A=\{1,2,3,4\}$ and $a R b$ if and only if $\mathbf{a \leq b}$

Determine whether the relation R is a partial order on the set A .

Solution:

$$R = \{(1,1),(1,2),(1,3),(1,4),(2,2), (2,3),(2,4),(3,3),(3,4),(4,4)\}$$

convert R to M_R $\longrightarrow$

$$M_R = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Solution

check:

a) **Reflexive** : have all 1's in its main diagonal

b) **Antisymmetric:** $M_{12}=1$ or $M_{21}=0$

$$M_{13}=1 \text{ or } M_{31}=0$$

$$M_{14}=1 \text{ or } M_{41}=0$$

$$M_{23}=1 \text{ or } M_{32}=0$$

$$M_{24}=1 \text{ or } M_{42}=0$$

$$M_{34}=1 \text{ or } M_{43}=0$$

c) **Transitive** : $(M_R)^2_{\odot} = M_R$

So, R is a partial order on the set A.

$$M_R = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

Hasse Diagrams

- A Hasse diagram is a type of mathematical diagram used to represent a partially ordered set (POSET).
- much simpler than its digraph
- The diagram is drawn vertically, with larger elements (greater in the order) placed higher.
- The Hasse diagram doesn't include any redundant or obvious connections.
- For example, if $A < B$ and $B < C$, the diagram skips the line between A and C because it's already obvious.
- The items higher in the diagram are "greater" than the ones below them.

Hasse Diagrams

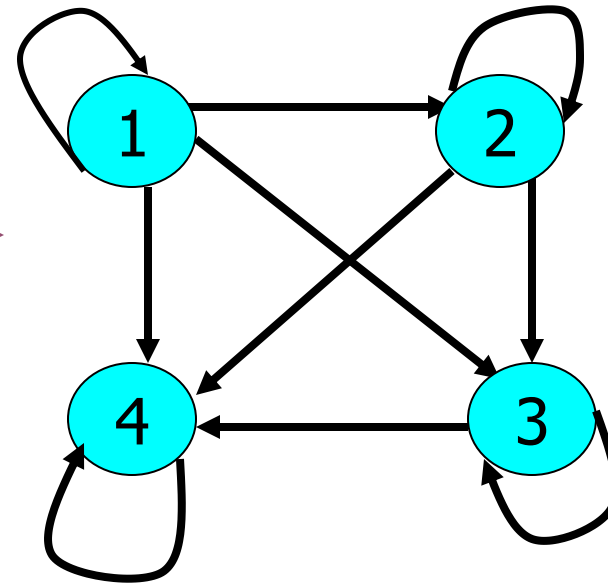
Steps in drawing the Hasse Diagram:

- 1.delete all cycles of length 1
- 2.eliminate all edges that implied by transitive property.
- 3.draw the digraph with all edges pointing upward.
- 4.replace circles with dots.

Using Example 1

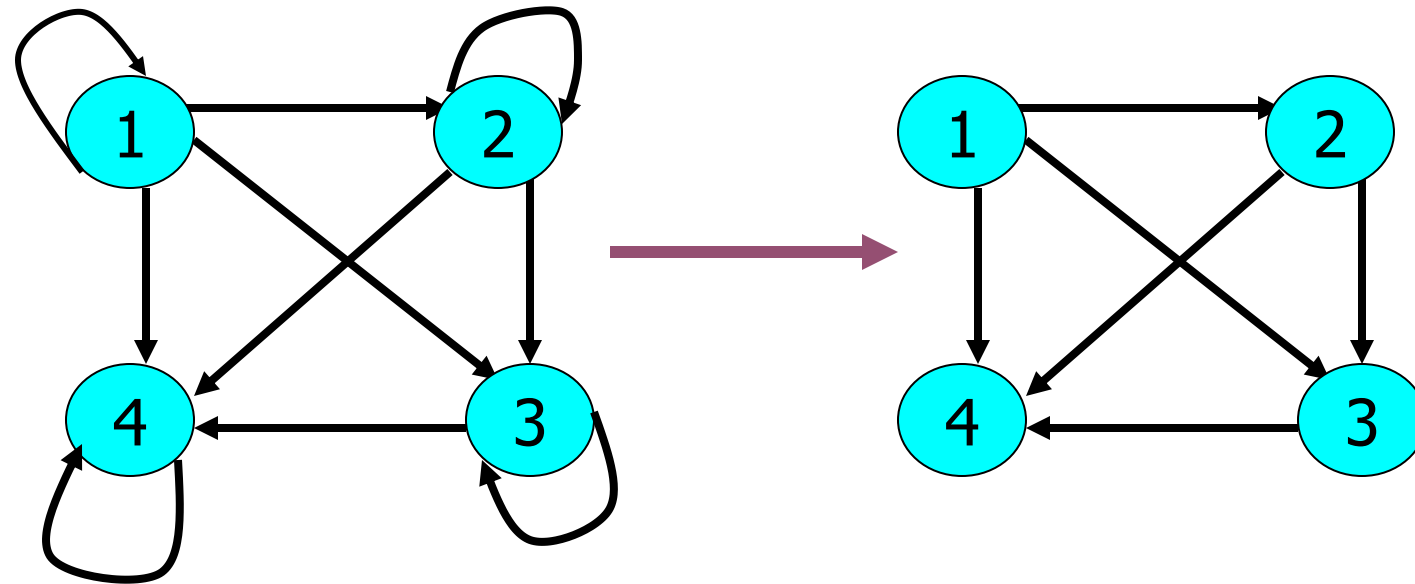
convert M_R to digraph R .

$$M_R = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$



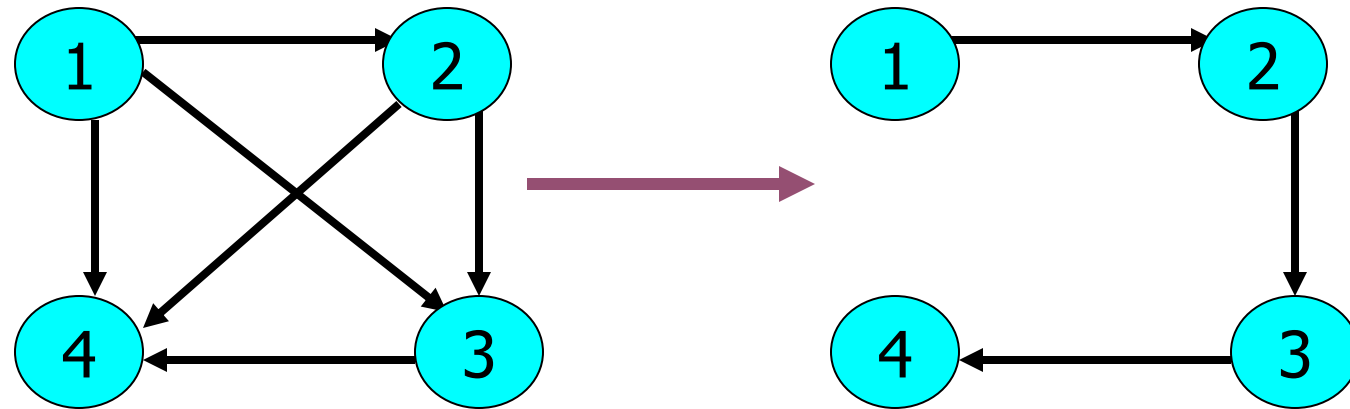
Using Example 1

Step1: delete all cycles of length 1



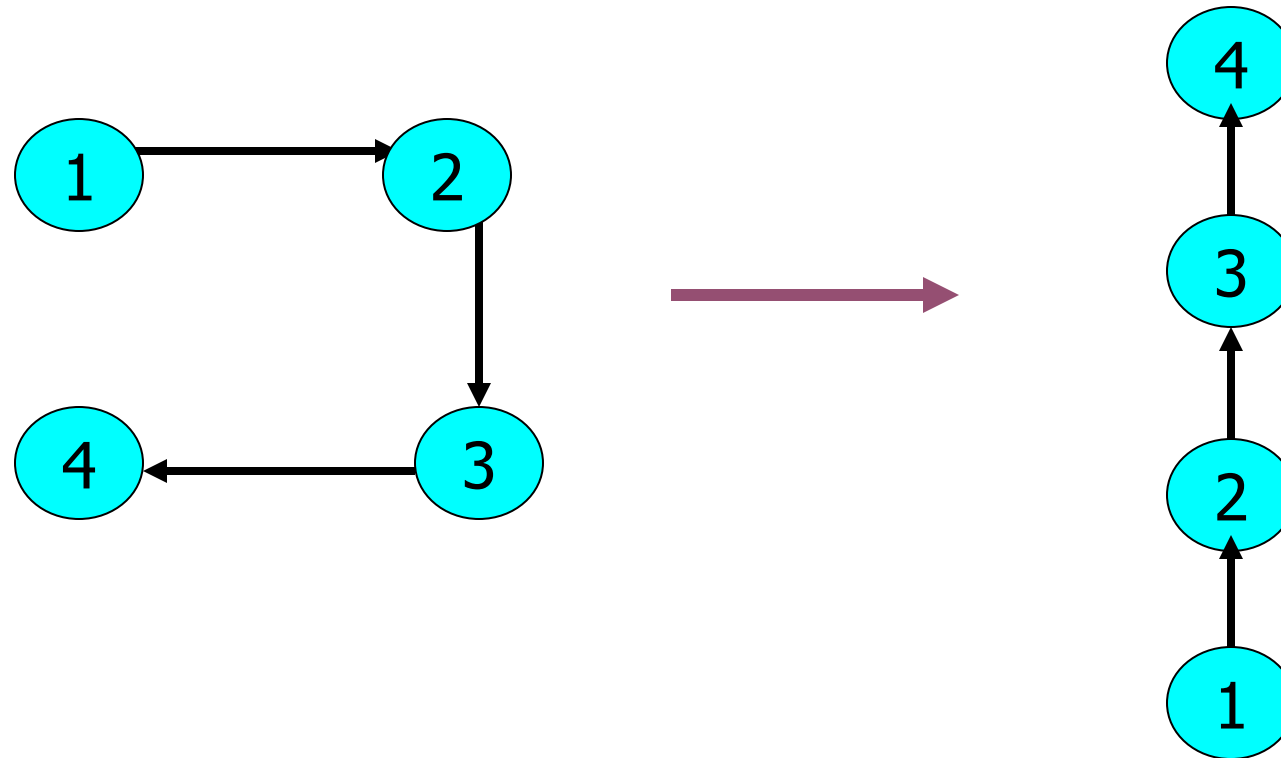
Using Example 1

Step 2: eliminate all edges that implied by transitive property.



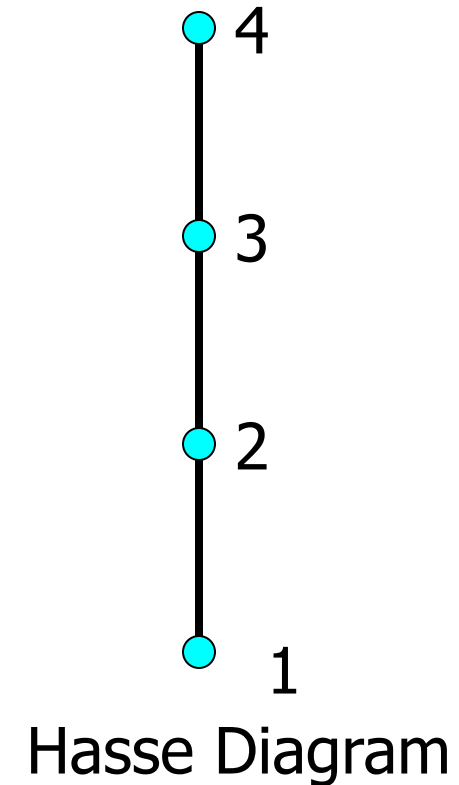
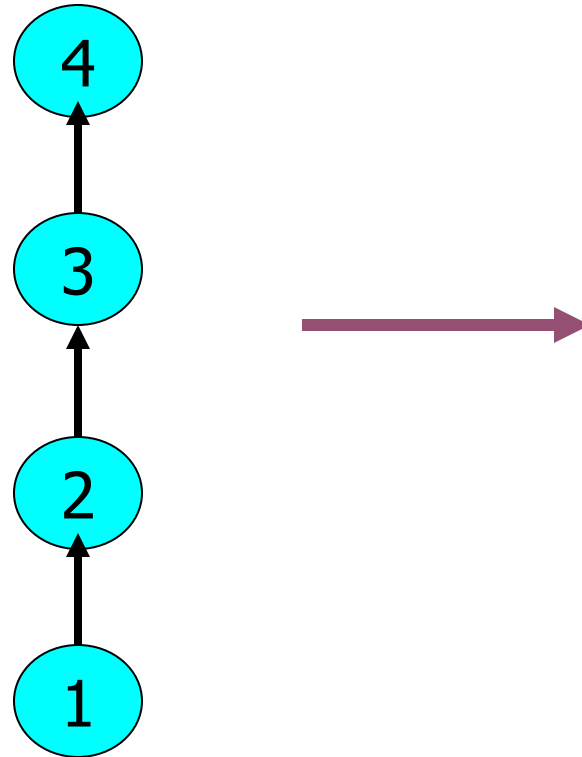
Using Example 1

Step 3: draw the digraph with all edges pointing upward.



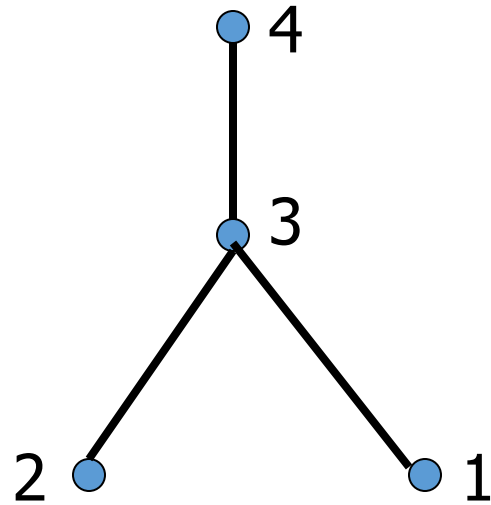
Using Example 1

Step 4: replace circles with dots



Example 2

Determine the matrix of the partial order whose Hasse diagram is given.



Solution:

$$R = \{(1,1), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$$

$$M_R = \begin{vmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix}$$